Reflection of Light

Exam code: 0625 & 0972

Download PDF

Written by: Katie M
Reviewed by: Caroline Carroll
Updated on 1 October 2024

Did this video help you? Yes No

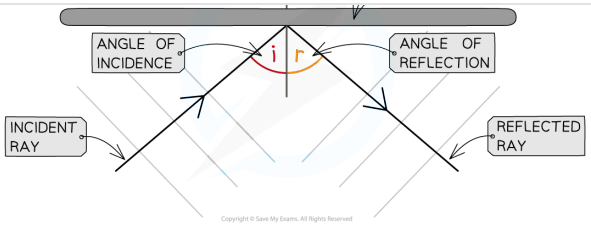
Ray diagrams

- In optics, a **normal line** is drawn at **right angles** to the **boundary** between two **media**
- In reflection, angles are measured between the ray (showing the wave direction) and the normal line
  - The angle of the wave approaching the boundary is called the **angle of incidence *(i)***
  - The angle of the wave leaving the boundary is called the **angle of reflection *(r)***
- When drawing a ray diagram an arrow is used to show the direction the wave is travelling
  - An **incident** ray has an arrow pointing **towards** the boundary
  - A **reflected** ray has an arrow pointing **away** from the boundary

Ray diagram of reflection

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



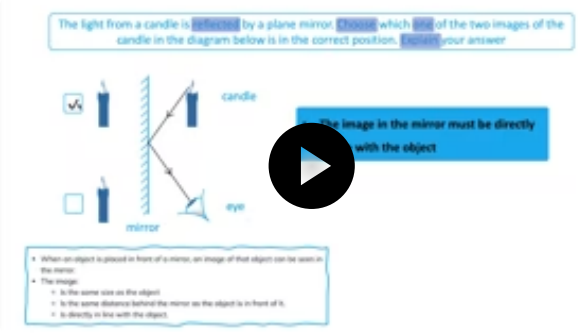
A ray diagram for light reflecting at a boundary, showing the normal, angle of incidence and angle of reflection

The law of reflection

- The **law of reflection** states that the angles of incidence and reflection are equal:

  Angle of incidence (*i*) = Angle of reflection (*r*)

- This law of reflection is used in many devices such as:
  - mirrors
  - cameras
  - optical fibres
  - periscopes



Did this video help you? Yes No

Reflection in a plane mirror

- When an object is placed in front of a plane mirror, an image of that object can be seen in the mirror
- The image will be:
  - The **same size** as the object
  - The **same distance behind** the mirror as the object is in front of it
  - **Virtual**

- A **plane mirror** defines a **flat, smooth** and **polished surface**

Ray diagram showing reflection in a plane mirror

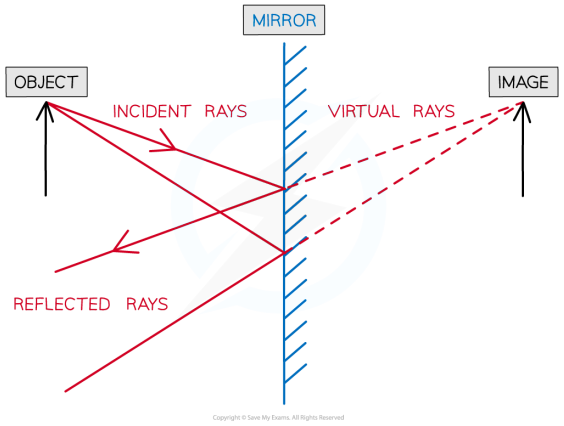


Diagram showing the formation of an image in a mirror by the reflection of light

- Each incident ray on the diagram above can be drawn following these steps:
  - Light from the object hits the mirror, reflecting from it ($i=r$)
  - To an observer, the reflected ray appears to have come from behind the mirror
  - The reflected ray can be traced back in this same direction behind the mirror, forming a **virtual ray**
  - This process is **repeated** for another ray travelling in a slightly different direction
- An image of the object will appear where these two **virtual rays cross**
- The type of image formed in the mirror is called a **virtual image** because of the divergence of the rays from the image
  - It cannot be projected onto a piece of paper (because the rays don't go through the image)

Examiner Tips and Tricks

When drawing light waves being reflected **take care to get the angle about right.**

If they are slightly out it won't be a problem, but if there is an obvious difference between the angle of incidence and the angle of reflection then you will probably lose a mark.

Teacher only

Fresh, expert-written questions
Students can't access them
Fair accurate tests - built fast
Explore in Test Builder
Test yourself
Flashcards
Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

Resources
Learning Hub
Scholarship
Join
Past Papers
Solution Banks
Sitemap

Dashboard
Overview
Brand ambassador
Profile
Subscription
Settings
Log out

Company
About us
Exam Specificity
Content Quality
Promotions
Jobs
Terms
Privacy
Cookie Policy
Help and Support

Subjects
Biology
Chemistry
Physics
Maths
Geography
English Literature
Psychology
All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.

Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.